

10 hour
 $12 \times 60 \times 60 = 43200$
 $43200 \times 12 = 518400$
 $518400 \times 12 = 6220800$
 $6220800 \div 2 = 3110400$

36
 $36 \times 12 = 432$

43200
 $43200 \div 12 = 3600$
 $3600 \div 6 = 600$

Instruction I: Choose the choice with the right answer and write the letter of your choice on the shaded space provided ONLY (1 points each)

- Let z_1 and z_2 be two different complex numbers and $z = (1-t)z_1 + tz_2$ for some real number t with $0 < t < 1$. If $\text{Arg } w$ denotes the principal argument a complex number w , then $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$ is
 - $\frac{\pi}{2}$
 - 0
 - $-\frac{\pi}{2}$
 - $\text{Arg } z = \text{Arg } z_2$
- If the eccentricity values of four hyperbolas with the same transverse axis length are e_1, e_2, e_3 and e_4 where $e_4 < e_3 < e_2 < e_1$, then the arms of the hyperbola open more in the hyperbola with eccentricity of
 - e_3
 - e_4
 - e_2
 - e_1
- The exact numerical value of $-36 \cosh(-\ln 3) - 36 \sinh(\ln 2)$ is _____.
 - 87
 - 52
 - 45
 - 66
- For propositions p, q and r , which compound proposition is not tautology?
 - $(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$
 - $(r \Rightarrow \neg q) \Rightarrow (q \Rightarrow \neg r)$
 - $\neg(\neg q \Rightarrow \neg p) \Leftrightarrow (q \wedge p)$
 - $\neg(r \Rightarrow \neg p) \Leftrightarrow (p \wedge r)$
- Which of the following is **not true** about the conic section $\frac{4}{489}(x^2 - 6x) - \frac{25}{489}(y^2 - 10y) = 1$?
 - Center is (3,5)
 - transverse length is 4
 - one focus is (3,9)
 - one vertex is (3,7)
- Which one has truth value false in the set of natural numbers?
 - $(\exists x)(\forall y): x \leq y$
 - $(\forall x)(\forall y): (x+y)^2 \geq 2xy$
 - $(\forall x)(\exists y): x = y^2$
 - $(\exists x)(\exists y): x^2 + y^2 = x + y$

Instruction II: Fill the following blank spaces by the most simplified answers on the given shaded spaces ONLY. (2 points each)

- The radius of a circle passing through the points (3,-1), (0,0) and (-1,-1) is 2
- Two lights are turned on at the same time. One blinks every four seconds and the other blinks every six seconds. A dog shouts every time the two lights blink together. The dog shouts 18000 times in 12 hours.
- For two sets A and B if $n(A \setminus B) = n(B \setminus A) = 80$ and $n(A \cap B) = 30$, $n(A \cup B) =$ 190
- The real part of $(1+i)^{20}$ after simplification is -2^{10}
- A polynomial p of degree six has zeros of -3 and 2 with multiplicities four and two respectively. If $p(-1) = -288$, then $p(-2) =$ -32
- If $-5+2i$ is one root of the equation $x^2 + kx + m = 0$, then the value of m is 29
- If $f(x) = \frac{e^x - e^{-x}}{2}$, then $f^{-1}(\sqrt{e^2 - 1}) =$?

$(c+2i)(c-2i) = m \quad f^{-1} =$

2. Use principle of mathematical induction to prove that
 $x-y$ is a factor of $x^n - y^n$, $\forall n \in \mathbb{N}$ = set of natural numbers

(6 points)

Step 1: For $n=1$, ~~$x^1 - y^1$~~ $x-y$ is a factor of $x^1 - y^1$
 $x-y$ is a factor of $x-y$ it is true

Step 2: Assume that it is true for $n=k$, where $k \in \mathbb{N}$
 i.e. $x-y$ is a factor of $x^k - y^k$ $\Rightarrow$ $x^k - y^k$ is divisible by $x-y$
 we can write in this form $x^k - y^k = m(x-y)$ where $m \in \mathbb{Z}$

Step 3: We should show that it is true for $n=k+1$

Claim: $x^{k+1} - y^{k+1}$ is divisible by $x-y$, ~~ble $x-y$ factor~~
 (Hint: Any number is divisible by its factor)

From Assumption step we have

$$x^k - y^k = m(x-y)$$

Now: $x^{k+1} - y^{k+1} = l(x-y)$, where $l \in \mathbb{Z}$

$$x^k \cdot x - y^k \cdot y$$

$$x^k \cdot x - x^k y - y^k y + y^k x + x^k y - y^k x$$

$$x^k(x-y) - y^k(x-y) + x^k y - y^k x$$

$$(x^k - y^k)(x-y) + x^k y - y^k x$$

$$m(x-y)(x-y) + x^k y - y^k x$$

12 hour

12 x 60 x 60
3600 x 12
00

x 36
12
2

x 36
12
172
36
432

43200
12
10800
12
7200 - 6

Instruction III: Work out the following questions showing all the necessary steps clearly on the given space provided (6 points each)

1. Consider the argument,

If Ethiopia is the country of thirteen months of sunshine, then Ethiopia is naturally endowed country.

It is not true to say that either Ethiopia is naturally endowed or Ethiopia is mountainous country.

Therefore, if Ethiopia is not the country of thirteen months of sunshine, then Ethiopia is not mountainous country.

- Write the above argument using logical operators (2 points)
- Check the validity of the argument using truth table (4 points)

let $P(x)$: Ethiopia is the country of thirteen months of sunshine

$q(x)$: Ethiopia is naturally endowed country

$r(x)$: Ethiopia is mountainous country

i. $P \Rightarrow q, \neg(q \vee r) \vdash \neg P \Rightarrow \neg r$

ii.

	p	q	r	$P \Rightarrow q$	$q \vee r$	$\neg(q \vee r)$	$\neg P$	$\neg r$	$\neg P \Rightarrow \neg r$
1	T	T	T	T	T	F	F	F	T
2	T	T	F	T	T	F	F	T	T
3	T	F	T	F	T	F	F	F	T
4	T	F	F	F	F	T	F	T	T
5	F	T	T	T	T	F	T	F	F
6	F	T	F	T	T	F	T	T	T
7	F	F	T	T	T	F	T	F	F
8	F	F	F	T	F	T	T	T	T



The argument has truth value True on 8th row



3. Given a rational function $f(x) = \frac{(3x^2 - 12)(x+1)}{x^2 - 2x - 3}$, find its
- domain (1 point)
 - x and y intercepts if any (1 point)
 - All possible existing asymptotes if any (2 points)
 - Sketch the graph (2 points)

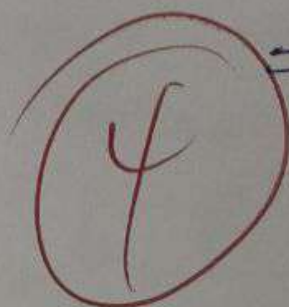
Soln

$$\begin{aligned} \text{a) } f(x) &= \frac{3(x^2 - 4)(x+1)}{(x-3)(x+1)} \\ &= \frac{(3)(x^2 - 4)}{x-3}, \quad x \neq -1 \end{aligned}$$

∴ It is domain the number that make denominator zero.
Hence, Domain $\mathbb{R} / \{-1, 3\}$

$$\text{b. } f(x) = \frac{(3x^2 - 12)(x+1)}{(x-3)(x+1)} = \frac{3x^2 - 12}{x-3}, \quad x \neq -1$$

$$x\text{-intercept, } y=0 \Rightarrow \frac{3x^2 - 12}{x-3} = 0,$$



$$\Rightarrow 3x^2 - 12 = 0,$$

$$3(x^2 - 4) = 0$$

$$(3)(x+2)(x-2) = 0$$

$$x+2=0 \text{ \& } x-2=0$$

$$x = -2 \text{ \& } x = 2$$

4. The vertices (end points of the major axis) of an ellipse $x^2 + 4y^2 - 4x - 24y + 24 = 0$ are the vertices of a hyperbola. If the point $(2 + \sqrt{32}, 6)$ lies on the hyperbola, Find

- the equation of the hyperbola? (2 points)
- the coordinates of the focal points of the hyperbola? (2 points)
- the asymptotes of the hyperbola (2 points)

Solution

$$x^2 - 4x + 4y^2 - 24y + 24 = 0$$

$$x^2 - 4x + 4 + 4(y^2 - 6y + 9) = 4 + 36 - 24$$

$$(x-2)^2 + 4(y-3)^2 = 16$$

$$\frac{(x-2)^2}{16} + \frac{(y-3)^2}{4} = 1 \quad \text{standard form of eqn of ellipse}$$

The ellipse is horizontal with centre $C(2, 3)$ & $a=4$

$V(x, y)$ & $V'(x', y')$

$$V(\overset{2+a}{\cancel{2+4}}, 3), \text{ but } a=4$$

$$V(4+2, 3) \text{ \& } V'(2-4, 3)$$

$$V(6, 3) \text{ \& } V'(-2, 3)$$

We get the vertices of hyperbola $V(6, 3)$ & $V'(-2, 3)$

the midpoint of two vertices is centre of hyperbola

So, $C(2, 3)$, the length of the major axis equals to the length of transverse axis

$$\text{i.e. } 2a = VV' \Rightarrow a=4$$

We know that the hyperbola also the transverse axis is parallel with x -axis so, the eqn becomes

$$\frac{(x-2)^2}{16} - \frac{(y-3)^2}{b^2} = 1, \text{ But the } P(2 + \sqrt{32}, 6) \text{ on the hyperbola}$$

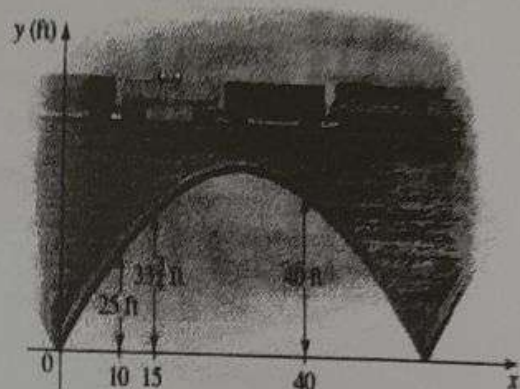
$$\frac{(2 + \sqrt{32} - 2)^2}{16} - \frac{(6-3)^2}{b^2} = 1, \text{ Solve for } b$$

$$12 \times 60 \times 60 \times \frac{36}{12} \quad | \quad \frac{36}{12} \quad | \quad \frac{43200}{12} = 3600$$

5. The opening of a railway bridge over a roadway is in the form of a tunnel (a parabola). A surveyor measures the heights of three points on the bridge taking xy axes with origin at 0, as shown in the figure below. He wishes to find an equation of

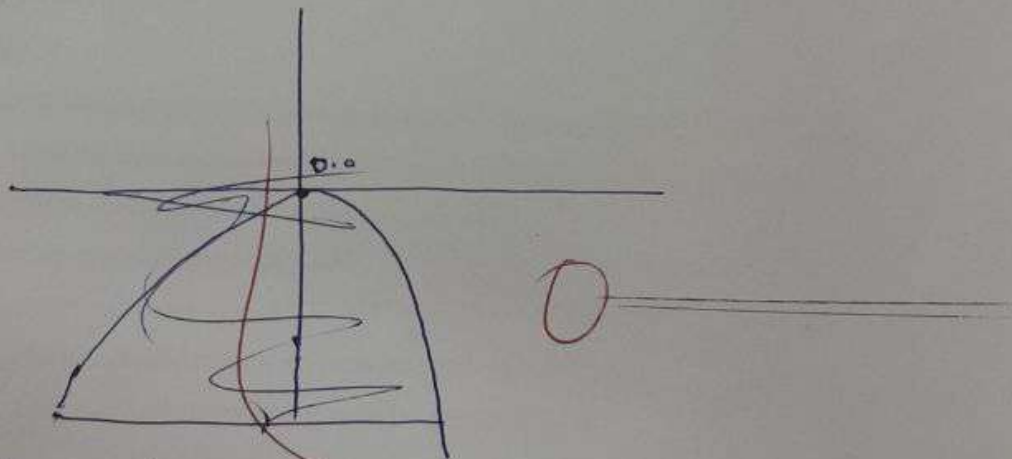
the form $\left(x + \frac{b}{2a}\right)^2 = \frac{1}{a}\left(y + \frac{b^2}{4a} - c\right)$, $a \neq 0$ to model the shape of the parabolic arch. Find

- The standard equation of the parabola after calculating the values of a, b and c (3 points)
- The focal length of the parabola (1 point)
- Is it possible for a lorry of height 45.25 ft to pass through the tunnel? (justify your answer) (2 points)



we know that $(10, 25)$, $(15, 33)$, $(40, 40)$ the point on the parabola

$$(x-h)^2 = -4p(y-k)$$



$$(10-h)^2 = -4p(25-k)$$

$$100 - 200h + h^2 = -4p(25-k)$$